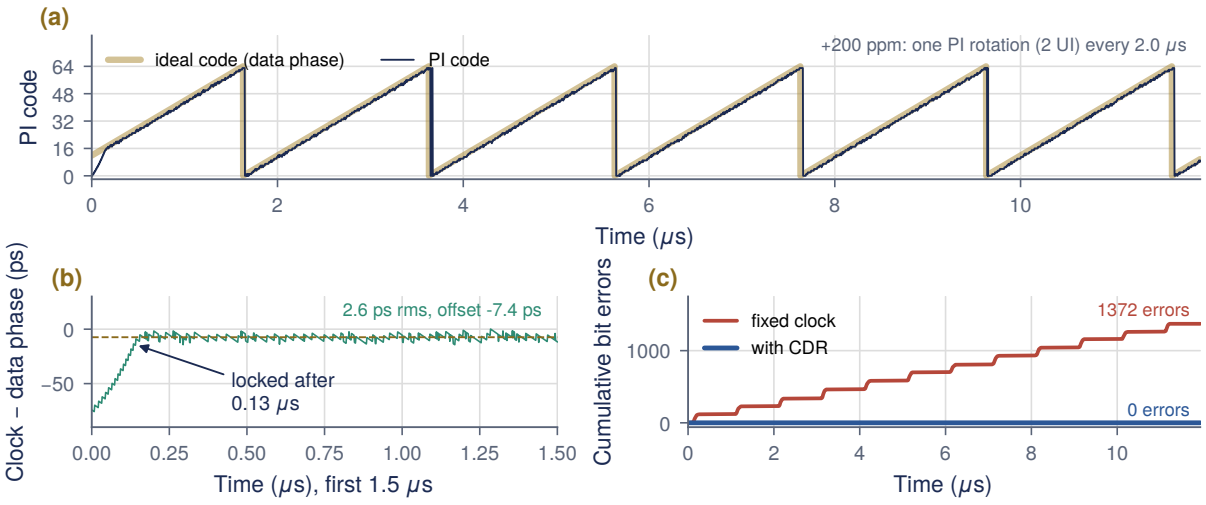
(a)
ideal code (data phase)
PI code
+200 ppm: one PI rotation (2 UI) every 2.0 μs
PI code
64
48
32
16
0
0
2
4
6
8
10
Time (μs)
(b)
2.6 ps rms, offset -7.4 ps
0
−50
locked after
0.13 μs
Clock – data phase (ps)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
Time (μs), first 1.5 μs
(c)
fixed clock
with CDR
1372 errors
0 errors
1000
0
Cumulative bit errors
0
2
4
6
8
10
Time (μs)